because these assets have illiquidity risk and cannot be continuously traded to construct an "arbitrage."

Investors Must Demand High Illiquidity Hurdle Rates

How much does an investor need to be compensated for illiquidity? In Panel B of Table 13.6, I compute premiums on an illiquid asset required by an investor to bear illiquidity risk. Let's define the illiquidity premium, or hurdle rate, as a certainty equivalent (see chapter 2). Suppose an investor holds two liquid assets and replaces one asset with another that is identical except for being illiquid. The illiquidity premium is the increase in the expected return of the illiquid asset so that the investor has the same utility as the case when all assets are liquid.

When liquidity events arrive every six months, on average, an investor should demand an extra 70 basis points. (Some hedge funds have lockups around this horizon.) When the illiquid asset can be traded once a year, on average, the illiquidity premium is approximately 1%. When you need to wait ten years, on average, to exit an investment, you should demand a 6% illiquidity premium. That is, investors should insist that private equity funds generate returns 6% greater than public markets to compensate for illiquidity. As Section 3 discusses, most illiquid assets are not generating excess returns above these hurdle rates (see also chapter 11).

The Ang, Papanikolaou, and Westerfield (2013) model is highly stylized. Given the other issues the model misses, like agency conflicts of interest (see chapter 15), cash flow management issues of capital calls and distributions, and asset–liability mismatches, the true illiquidity hurdle rates are even higher than those reported in Table 13.6.

5.3. SUMMARY

Portfolio choice models with illiquid assets recommend holding only modest amounts of illiquid assets. Investors should demand high illiquidity risk premiums.

6. Liquidating Harvard Redux

6.1. THE CASE FOR ILLIQUID ASSET INVESTING

Large, long-term investors often cite their large amounts of capital and their long horizons as rationales for investing in illiquid assets. Size and patience are necessary but not sufficient conditions for illiquid asset investing; these conditions simply aren't adequate justifications in themselves. Since illiquid asset classes do not offer high risk-adjusted returns, the case for passively them is not compelling. Illiquid investing also poses huge agency problems; asset owners, for example, find